Lunar Life: Progress Report

NASA Space Apps 2026 | Prepared 14 November 2026 for Jimmy | From Aryan

Summary

Lunar Life is a cozy 16-bit astronaut life sim for kids, built in Godot 4.7. Players train at an academy, then run a lunar outpost where real NASA space-weather alerts force power and safety trade-offs. We are building one polished 6-minute slice. So far the design plan, the power simulation, the first three steps of the slice and the four main character sprites are done. The NASA data, the outpost and the alert event are still to build.

What has been done

- **Design and technical plan.** A full plan doc: the 7-step slice, core loop, resource model, Godot architecture, art rules, NASA data sources, 48-hour roadmap and science-accuracy notes.
- **Godot project set up.** Pixel-art settings locked: 480 x 270 base resolution, integer scaling, nearest filtering, pixel snapping.
- **Core systems.** EventBus (shared signals), GameState (name, suit colour, scores, certifications) and ResourceSim (solar power, battery, six loads by priority, automatic load shedding, life support, crops, science, radiation exposure in game units).
- **Power Console.** Working grey-box screen: toggle loads, watch the battery and a 12-hour sunlight forecast, speed controls and a Back button.
- **Create your astronaut.** Name entry and seven suit colours with a live recoloured preview.
- **Academy hub.** Walkable room (placeholder rectangles for now), three mentors with dialogue, and Engineer Rivera opens the Power Console training.
- **Supporting code.** Scene changes with a fade, a dialogue box, keyboard controls (WASD or arrows, E to talk) and a suit-recolour shader.
- **Art.** Four 32 x 32 character sprites made with SpriteCook: player astronaut, commander, engineer, botanist. 24 of 40 free credits used, 16 left.

The four main characters, enlarged 8 times.

Status against the 7-step slice

#	Slice step	Status
1	Create astronaut (name and suit colour)	Built
2	Academy hub and 3 mentors	Built; room art is placeholder
3	Power Console training	Console works as a sandbox; no goal, pass or certification yet

#	Slice step	Status
4	Graduate and arrive at the lunar outpost	Not started
5	Real NASA solar-particle (DONKI) alert	Not started
6	Three costed choices	Not started
7	Consequences, debrief, Science Library	Not started

What is left, in proposed order

- Give the training a goal and award the certification.
- Outpost room and the graduation transition.
- NASA data: save 3 to 5 real past DONKI solar-particle events as a fallback file, then add the live fetch.
- The alert event: three choices, their effects on power, crops, science and exposure, and the debrief.
- Science Library with 4 to 6 sourced entries.
- Art and sound pass: backgrounds, console screen, icons, audio.

Things to know

- None of the game code is committed to git yet. Only the initial commit exists.
- The player sprite has one front-facing pose, so walking is a bob and a left/right flip.
- Mentor names and dialogue are first drafts. They live in data/npcs/ and can be edited in Godot's Inspector.
- All simulation numbers are simplified game units, and the game says so on screen.
- Still to confirm: the local event's rules on code written before the start, and the exact DONKI API addresses, which moved on 30 September 2026.

Where things are

Item	Location
Design and technical plan	https://claude.ai/artifact/RzEQPuU9ESEESnCJM4ZNqh (private to Aryan until shared)
Godot project	nasahackathon folder on Aryan's Mac
Character sprites	assets/art/characters/
SpriteCook asset IDs	spritecook-assets.json